Intersections and faces in graph covers

Human Compression project

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Abstract

The fiber argument for simultaneous group actions also applies to an arbitrary equivariant projection. For a rank-four cover whose endpoint faces stay connected and whose middle faces lift isomorphically, this isolates the entire obstruction to polytopality in the lifted vertex–facet sections. We prove the face and thinness assertions explicitly. The application is the family of Leemans and Toledo; the covering-space and group-action facts are standard, and no historical priority is claimed for their reformulation here.

1 An equivariant projection

Let a group W act on sets Q and X, and let $\pi : Q \to X$ commute with the action. Fix $q \in Q$, and put $x = \pi(q)$. For $S \le W$, write $D_S(q) = Sq \cap \pi^{-1}(x)$. Thus

$$D_S(q) = (S \cap \mathrm{Stab}_W(x))q.$$

Lemma 1. *Suppose $C \le A \cap B$ and $Ax \cap Bx = Cx$. Then*

$$Aq \cap Bq = Cq \quad \Longleftrightarrow \quad D_A(q) \cap D_B(q) = D_C(q).$$

Proof. The forward implication restricts the asserted equality to the fiber of x. Conversely take $q' \in Aq \cap Bq$. Its image is cx for some $c \in C$. Then $c^{-1}q'$ lies in both orbits and in the fiber of x. The fiber hypothesis puts it in Cq, and hence $q' \in Cq$. The opposite inclusion is automatic. This is exactly the first-coordinate repair argument in *Intersections in simultaneous group actions*, Theorem 1: alternatively apply that theorem to the invariant graph $\{(\pi(q), q) : q \in Q\}$. □

For completeness, let Γ be a connected graph with a voltage $z(u, v)$ in a group K, satisfying $z(v, u) = z(u, v)^{-1}$. The cover has vertices (u, k) and edges from (u, k) to $(v, z(u, v)k)$. Products along walks are in traversal order on the left: the last edge acts last and therefore appears on the left of the product. If Y is a connected subgraph and $u \in Y$, the voltages of its closed walks at u form a subgroup $H_Y \le K$. Path lifting shows that the component of (u, k) in the preimage of Y meets the fiber of u in $H_Y k$. Every component meets that fiber, so its components are indexed by these cosets. In particular its preimage is connected exactly when $H_Y = K$, and each component maps isomorphically to Y exactly when $H_Y = 1$. These are the usual elementary voltage-cover facts, proved here by following paths; no finite enumeration is involved. If the voltage is trivial on a spanning tree, every non-tree dart completes a fundamental closed walk whose voltage is that dart's voltage. These fundamental walks generate all closed-walk voltages, giving the source's reduced-voltage connectivity criterion. For a cover of a maniplex, a distant-color square closes upstairs exactly when its four dart voltages multiply to the identity. Consequently the cover is a maniplex exactly when it is connected and every such square has trivial voltage. This also recovers source Lemmas 3.3 and 3.4.

2 A rank-four face criterion

In the geometric sections all graphs are finite, as in the source. A maniplex is a connected properly colored simple graph with one edge of every color at each vertex, and commuting nonconsecutive color adjacencies. Its flags are its vertices. An i-face is a component after deleting color i; incidence means nonempty intersection. A cover preserves colors and their unique adjacencies. Its restriction to any component of a prescribed set of colors is a cover of the corresponding component of the base: every color path lifts uniquely. All intersections below retain the flags and edges belonging to every intersected face. In a faithful maniplex this is also the induced graph on the common flags.

Faithfulness means that a flag is determined by all its faces. Thinness means that every interval between incident faces whose ranks differ by two has precisely two intermediate faces; improper endpoint faces are included. We recall the elementary chain realization behind the face-poset convention. Suppose a flag x is already in prescribed faces of ranks at least k, and we want also to meet an incident face F_j of rank $j < k$. Choose $y \in F_j \cap F_k$ and a path from x to y in F_k. Its colors avoid k. Every color below k commutes with every color above k, so reorder the path into its lower-color part followed by its upper-color part. The intermediate flag still belongs to all the prescribed higher faces; it also belongs to F_j, since the final upper-color part cannot change that face. Descending induction realizes every chain by a flag. The same argument for three ranks proves transitivity of the face order.

If the maniplex is faithful and thin, this realization is bijective for maximal chains and carries i-adjacency to changing just the rank-i face. Indeed i-adjacent flags lie in all the same other faces, and their i-faces differ by faithfulness; thinness says these are the only two completions. Thus a faithful thin maniplex is polytopal exactly when all face-chain intersections are connected. This is the usual connected- intersection characterization of Garza-Vargas and Hubard; the preceding argument gives precisely the direction and hypotheses used here.

Lemma 2. *Every faithful thin rank-three maniplex is polytopal. In a faithful rank-four maniplex each middle face is a prism: a single edge times an even alternating polygon. If the rank-four maniplex is also thin, every chain containing a middle face has connected intersection.*

Proof. In any faithful maniplex, i-adjacent flags cannot belong to the same i-face: their other faces already agree, and agreement of the remaining face would contradict faithfulness. For a faithful thin rank-three maniplex, an intersection of two faces consists of the two flags completing that chain, joined by the remaining color. Three faces have one common flag, a singleton face is connected by definition, and the whole graph is connected. The characterization just proved applies. This recovers the full scope of Leemans–Toledo Lemma 2.6.

For a rank-four 1-face, consider one colors-2, 3 polygon. Color 0 commutes with both of these colors, so it maps this polygon isomorphically to another. The polygons are distinct: the first is contained in one 0-face, and a 0-neighbor cannot remain in that face. Connectivity then says these two polygons exhaust the 1-face, paired by color 0. This proves the prism assertion; the 2-face is dual. In particular a 1-face meets exactly two 0-faces, and a 2-face meets exactly two 3-faces. No thinness assumption was used here.

Assume now thinness. A chain of at least three proper faces has either one flag or the two flags of the missing-color edge, as above. For a pair of faces it remains to examine five pairs of ranks. An incident $0, 1$ pair intersects in exactly one of the polygon layers just described; the $2, 3$ pair is dual. For a $1, 2$ pair there are at most two choices of vertex and two of facet, hence at most four flags by faithfulness. The colors-$0, 3$ square through any common flag supplies four, so is the whole intersection. For a $0, 2$ pair thinness gives exactly two possible intermediate 1-faces, and the prism description gives two possible facets. Again there are at most four flags, and the colors-$1, 3$ square supplies all four. The $1, 3$ case is dual. These are all pairs containing a middle face. Singleton faces and the larger chains already considered finish the proof. □

Theorem 3. *Let P be a faithful thin rank-four maniplex, and let $\pi : Q \to P$ be a connected maniplex cover. Assume:*

1. *the full preimage of every* 0*-face and every* 3*-face is connected;*
2. *every component over a* 1*-face or a* 2*-face maps isomorphically onto that face.*

Then Q is faithful and thin. Every chain containing a middle face has a connected nonempty intersection, mapped isomorphically to the intersection of its image faces. Consequently Q is polytopal if and only if the preimage of every incident vertex–facet intersection in P is connected.

Proof. Endpoint faces of Q are the full preimages of endpoint faces of P. Fix middle faces H_1, H_2 of Q that meet, and write $F_i = \pi(H_i)$. The base intersection $F_1 \cap F_2$ is connected. Starting at a flag of $H_1 \cap H_2$, every path in that base intersection lifts to a path in both faces, because it uses neither color 1 nor color 2. Since each H_i maps isomorphically, there is exactly one lift of each base flag in either H_i. Their restrictions therefore agree over the entire intersection. Thus $H_1 \cap H_2$ maps isomorphically onto $F_1 \cap F_2$. This establishes the projection-of-intersection equality in this specific situation; such equality is not assumed for arbitrary covers.

Now consider any chain containing at least one middle face. Its endpoint faces impose exactly the conditions of belonging to their full preimages. If both middle faces occur, the preceding paragraph identifies their intersection; if only one occurs, its own projection is an isomorphism. In either case restricting that isomorphism to the endpoint conditions identifies the whole intersection with its base-chain intersection. The latter is connected by Lemma 2 and nonempty by chain realization. For a maximal chain it is a singleton. In particular each flag is determined by its incident faces, which is faithfulness.

We verify thinness instead of inferring it just from injectivity of a projection. Fix a chain missing one rank i. It still contains at least one middle face. Its intersection is, by the preceding argument, isomorphic to the base intersection for the image chain. That base intersection consists of precisely the two flags of an i-edge. Both lifted flags are in all the prescribed faces. Their respective i-faces complete the chain, and these faces are distinct, since otherwise faithfulness would identify the two flags. Conversely, every completion has a flag in its full intersection by the maximal-chain argument, so its i-face is one of these two. Every codimension-one chain therefore has precisely two completions. Extending any incident pair of ranks $i-1, i+1$ to such a chain shows the diamond condition: any intermediate i-face is comparable with every chosen lower or upper face by transitivity of the face order, and hence is one of those two.

The only remaining chains have no middle face. A singleton face is connected; a chain consisting of a vertex and a facet has intersection exactly the full preimage of their base intersection. The empty chain has intersection Q, which is connected. The connected-intersection characterization with its faithful and thin hypotheses proves the last assertion. □

Corollary 4. *Let P be a faithful rank-four maniplex whose* 0*- and* 3*-faces are nonbipartite. Its canonical double cover, with adjacency $(u, \epsilon)^i = (u^i, \epsilon + 1)$, is connected and faithful and fails the connected-intersection property. It has the same numbers of vertices and facets and twice the numbers of edges and two-faces. If P is thin, so is its double cover, which is then nonpolytopal. If P is regular, so is its double cover, and its automorphism group is* $\mathrm{Aut}(P) \times C_2$.

Proof. Along a lifted walk the second coordinate records its length modulo2. The double cover of a connected graph is therefore connected precisely when that graph has an odd closed walk; otherwise it consists of two isomorphic copies. Endpoint faces have connected preimages by hypothesis. The middle faces are bipartite prisms by Lemma 2, so each has two isomorphic lifts. This proves connectedness and the face counts. If two flags upstairs lie in all the same faces, their projections do so in P and hence coincide. Both upstairs flags are in one lifted middle face, where projection is injective, so they

coincide as well. This proves faithfulness without a thinness assumption. When P is thin, Theorem 3 proves thinness of the cover.

Choose incident endpoint faces in P. Their intersection has only colors 1,2, with one edge of each color at every flag. It is a disjoint union of even alternating polygons. If there is more than one component, its preimage is disconnected; if there is one, its canonical double already has two components. Since full endpoint preimages are faces, this disconnected preimage is an actual face intersection upstairs. Thus the double cover fails the connected-intersection property. If it is thin, its proved faithfulness makes it nonpolytopal.

Each automorphism α of P acts by $(u, \epsilon) \mapsto (\alpha(u), \epsilon)$, commuting with the deck flip. If P is regular, this gives a transitive subgroup $\mathrm{Aut}(P)\times C_2$. A color-preserving automorphism of a connected maniplex is determined by one flag, so the transitive semiregular subgroup is the entire automorphism group. This proves every clause of source Lemma 3.8 on its stated faithful base, despite the more general thin-base Lemma 3.5 being false. □

3 Lifting an automorphism

The standard sufficient lift condition has a useful version with a change of coordinates in each fiber. Suppose voltages take values in an additive abelian group K, ϕ is a base automorphism, $f \in \mathrm{Aut}(K)$, and h is a function on the base vertices. If every dart uv satisfies

$$z(\phi u, \phi v) = f(z(u, v)) + h(v) - h(u),$$

then $(u, k) \mapsto (\phi u, f(k) + h(u))$ is a lift of ϕ. It is bijective on each fiber; the displayed equation says exactly that the images of adjacent vertices are adjacent. This proves the claim. For arbitrary multiplicative K and $h = 0$, the same direct calculation proves $(u, k) \mapsto (\phi u, f(k))$ whenever $z(\phi u, \phi v) = f(z(u, v))$. This is source Lemma 5.5, attributed there to Malnič–Nedela–Škoviera and Potočnik–Toledo. We use only this elementary sufficient condition, not an unstated converse.

4 The necessary restriction on the base

The faithful hypothesis in Theorem 3 cannot simply be dropped. Leemans–Toledo Lemma 3.5 assumes only a thin base with the same two conditions on face preimages. The application gives a complete explicit 32-to-64-flag counterexample. Its base is thin but unfaithful; its cover is faithful but not thin. Thus the exact general statement is disposed of by a counterexample, while Lemma 2 and Theorem 3 supply a sufficient repair covering the source's faithful construction. No claim about the source's principal theorem is inferred from failure of that auxiliary lemma.